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One Page Decisions Document:

First on AI tools, I used claud ai to help write the majority of the code. This was especially useful for understanding libraries such as open CV that I had not used before and for making figures. Without ai coding tools these steps would have taken alot longer to look up and read documentation for. Another step that this was helpful was for cleaning up code in to nicer function and catching syntax errors; both can be very time-consuming steps. For any measurement numbers produced by code written by either ai or me, it is important to include verifications and checks against standards or known values wherever possible.

For the calculation of the pixel pitch in mm, we can calculate it two ways. One from the calibration using the checkerboard and again from the crop factor of the camera that is known, and compare the percent difference between the two. My difference was 2.62774%. This shows a good agreement between the two. Checking that our calibrated values could be used to undistort a checkerboard near an edge was a nice visual that the process is working.

To validate the code written to measure the MTF from a slanted edge we generated a synthetic edge with a known gaussian noise added. Then measure the MTF and MTF50 on the synthetic edge using code. This allows us to check the agreement of our code against a known analytic solution.

For the calculation of the photon transfer function, we can verify our code using generated camera sensor data white images with a known gain and read noise. Then we can calculte the gain and read noise using our photon transfer function code and compare the result to the known values. Our code produces a percent error for the gain (-0.69%), and a percent error for the read noise (5.72%).

The manufacturer also includes noise model numbers with the captures; this gives us another value to compare our results to. Our K gain was 0.4099 ± 0.0038 DN/e- which was larger than the manufacturer value of 0.1014 DN/e-. This then results in a smaller read noise value for our measurements than the manufacture. I guess this surprised me as I was feeling good about mine, given we had tested our code on the simulated data and comparing the calculated and true values for K and read gave small percent error values. I believe that this means that our code is not likely the issue here. This means the discrepancy would be from either my captures or differences/errors in the vendor calculation. Possible errors in my captures: one would be maybe our flat white fields are unevenly illuminated, or maybe there was movement between the pairs of captures increasing the variance too much, or maybe I had different iso between the captures, or maybe some pairs do not have matching apertures. The ISO and aperture concerns as well as any concerns of not using the correct camera should be caught by checks in the code

above, so I rule this possibility out. The other two problems should be mitigated by the fact that I use only a middle patch of the whit field illumination so we should not be effected by vignetting and that I used a sort of homemade stand (out of books) to keep the camera in the same place while measuring (however this is relatively crude set up so it could certainty not be perfectly stable). Looking into how good our fit is; plotting the residuals they seem to be randomly around 0. This is a good sign that my fit is ok.

For possible assumptions of the manufacture result that may account for the difference include, we can't confirm that theirs is for the same iso or different gain than us, maybe they do other processing on their images before calculation the fit, and they also do not report an uncertainty on their measurements.